Fiscal Year:	FY 2018	**Task Last Updated:**	FY 11/02/2017
PI Name:	Zea, Luis Ph.D.		
Project Title:	Characterization of Biofilm Formation, Growth, and Gene Expression on Different Materials and Environmental Conditions in Microgravity		
Division Name:	Physical Sciences, Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Microbiology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	80309-0429	**Congressional District:**	2
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2015 NNH15ZTT002N MaterialsLab Open Science Campaigns for Experiments on the International Space Station
Start Date:	01/03/2017	**End Date:**	01/02/2020
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA MSFC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Klaus, David Ph.D. (University of Colorado, Boulder) Moeller, Ralf Ph.D. (Deutsches Zentrum Fuer Luft- Und Raumfahrt E.V.) Muecklich, Frank Ph.D. (Saarland University) Stodieck, Louis Ph.D. (University of Colorado, Boulder)		
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Task Description:	Biofilm formation and its concomitant risk of causing equipment malfunction and human illnesses is a problem that needs to be addressed to enable safe long-term human space exploration. With that objective, this proposal aims to characterize biofilm growth during one International Space Station (ISS) increment in terms of mass, thickness, morphology, and associated gene expression using different spaceflight-relevant microbial species and material substrata. This project also aims to elucidate the biomechanical and transcriptomic mechanisms involved in the formation of the “column-and-canopy” biofilm architecture observed in space. The experiment is designed to assess the expression of genes associated with conferring microorganisms with resistance to oxidative stress, acidity, and antimicrobials in biofilms formed in space, and to determine physical mechanisms of material/microorganisms interaction in biofilms. The experiment will be conducted using BioServe's BioCells housing substrata coupons exposed to different microorganisms and media. Post-flight analyses of fixed samples include confocal laser scanning microscopy and RNA-sequencing. The significance of this project stems from gaining valuable new knowledge that may lead to improved methods and/or material properties for controlling biofilm formation during long duration space missions. The selection of bacterial species, growth media, and substrata material make the results applicable to a) the safety and maintenance of crewed spacecraft, b) planetary protection, c) mitigation of biofilm-associated illnesses on the crew, and d) biofilm-associated illnesses on Earth, thus making it relevant not only to this Research Announcement, but also to NASA’s Human Exploration and Operations Directorate, Space Biology Program, and Human Research Roadmap.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Biofilms are associated with multiple illnesses, including endocarditis (bacterial infection of cardiac tissue), cystitis (a urinary tract infection), and otitis media (an inflammatory disease of the middle ear), as well as several other hospital-acquired infections. Understanding how these biofilms form and their relationship with resistance to antibiotics can help us better prepare novel solutions to these problems on Earth.
Task Progress:	Two microbial organisms will be used in this experiment: a bacterial and a fungal model. Progress during the first year has been focused on defining the experiment protocol to be used in space, and is here reported in a per-organism basis. PSEUDOMONAS AERUGINOSA The bacterial model planned to be used in this experiment is P. aeruginosa. A scientific question that needs answering is what is the appropriate growth medium to be used to enable biofilm formation on the different material substrata planned to be used. For this, the following media have been tested: • 10% Tryptic Soy Broth (TSB); • 3% TSB ; • LB Broth; • LB Broth supplemented with 5 g/L glucose; • Modified Artificial Urine Medium; • 8.8 mg/L iodine solution; • Water On the following materials: • Cellulose membrane; • Aluminum 6061; • Titanium Allow Ti-6Al-4V; • Polycarbonate; • Quartz; • Silicone; • Stainless Steel 316; • Carbon Fiber Growth dynamics have also been assessed in each growth medium, without coupons, to serve as baseline data. Based on the preliminary results, media have been down-selected to two options: 10% TSB and LB Broth. PENICILLIUM CHRYSOGENUM The fungal model planned to be used in this experiment is Penicillium Chrysogenum (a.k.a. Penicillium Rubens). A scientific question that needs answering is what is the appropriate growth medium to be used to enable biofilm formation on the different material substrata planned to be used. For this, the following media have been tested: • Potato Dextrose Broth (PDB)'; • PDB supplemented with 5 g/L glucose; • 10% Tryptic Soy Broth (TSB); • 3% TSB; • 8.8 mg/L iodine solution; • Water On the same materials described for the bacterial organism as well as without coupons to characterize growth dynamics. Based on the preliminary results, the medium planned to be used for the spaceflight experiment is PDB. FIXATION AND MICROSCOPY A scientific question being answered with preliminary tests is what is the appropriate fixation protocol that is compatible with the constraints inherent with spaceflight and that will still enable post-flight microscopy data acquisition. Thus far, for phenotypic data acquisition, paraformaldehyde (PFA) at two different concentrations has been tested (1.5% and 4%). Stains that have been tested include DAPI, calcofluor white, acridine orange, and propidium iodide. Confocal and Fluorescent microscopy tests have taken place both at the University of Colorado Medical Campus as well as the Microbiology, Cellular, and Developmental Biology Facilities. PROTOCOL TESTING - Long-Term Media Stowage Tests. A spaceflight-derived constraint that usually does not exist on ground laboratories is that there may be a period of time between when the samples are prepared and when the experiment can start. This duration depends on multiple aspects, including potential launch delays and crew availability (for example, it can be months between the moment the media and samples are prepared on the ground prior to launch, and when the astronauts can go ahead and initiate the tests). Thus, the viability of the media and inoculum needs to be characterized as a function of time and stowage temperature. With this goal, the media described above has been stored at 4C and -20C (to enable flexibility in regards to temperature) for the last four months. This test is planned to continue up to six months, when growth dynamics will be characterized. DLIP MATERIAL COUPONS. Polycarbonate, carbon fiber, and silicone 1cm2 coupons have been sent to Co-Investigator (Co-I) Dr. Muecklich in Germany for DLIP (laser topography) processes. These material coupons are expected to be sent to Co-I Dr. Moeller in DLR and PI Dr. Zea early in the second year of the project. TESTS IN SPACEFLIGHT HARDWARE. The spaceflight hardware, BioServe’s 12-Well BioCell, was made available during the second half of the first year of the project. The BioCell has two gas-permeable membranes. Material coupons will be adhered to the inside of one of these two membranes.

	- Biofilm Formation on BioCell's Gas-Permeable Membrane. Tests have been performed to assess if biofilm forms on the PTFE membrane that is planned to be used in the spaceflight hardware, as we aim to have biofilm formation (cell adhesion to a material) to start on the material being tested, and not on the gas-permeable PTFE membrane and then migrate onto the material coupon. The results of these tests indicate that neither the bacterial or fungal strain form biofilm on the gas-permeable PTFE membrane in any media, except for the bacterial organism in LB Broth supplemented with 5 g/L glucose. - Coupon Adhesion. Tests have been performed to verify if the material coupons can be adhered to the inside of the BioCell (on one of the two gas-permeable membranes) and withstand launch loads. A two-fold approach has been taken for this: (1) operationally, a launch orientation requirement will be put in place for the coupons to be on the lower side of the BioCell, perpendicular to the gravity vector, and (2) tests in liquid medium and air have been conducted where the coupons are placed in the film facing down (i.e., a 1g tests, which is more stringent than the ug environment of spaceflight). Results confirmed that the two-sided tape planned to be used fulfills these requirements. - Adhesive biocompatibility. Tests were performed in the different media and organisms to assess if the use of the two-sided tape could have a confounding result on growth dynamics of biofilm formation. The results indicate that the selected tape does not influence growth dynamics or biofilm formation and thus can be used in the experiment. - Role of Gas Permeation on Growth. The BioCell has gas-permeable membranes that enable O2 and CO2 exchange. However, this gas exchange rate is not as high as what would exist if there were no membranes at all. Tests have been performed to characterize growth dynamics on both strains, in all the media mentioned before, and compared to no-membrane scenario. The results indicate that growth is still appropriate for testing and informs how long should the experiment last to ensure biofilm formation given the reduce gas exchange rates.
Bibliography Type:	Description: (Last Updated: 11/13/2025)
Abstracts for Journals and Proceedings	Cortesao M, Luo J, Moeller D, Nisar Z, Muecklich F, Hemmersbach R, Hellweg C, Zea L, Moeller R. "Growth and biofilm formation of Penicillium chrysogenum in simulated microgravity." 33rd Annual Meeting of the American Society for Gravitational and Space Research, Seattle, WA, October 25-28, 2017. 33rd Annual Meeting of the American Society for Gravitational and Space Research, Seattle, WA, October 25-28, 2017. , Oct-2017
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